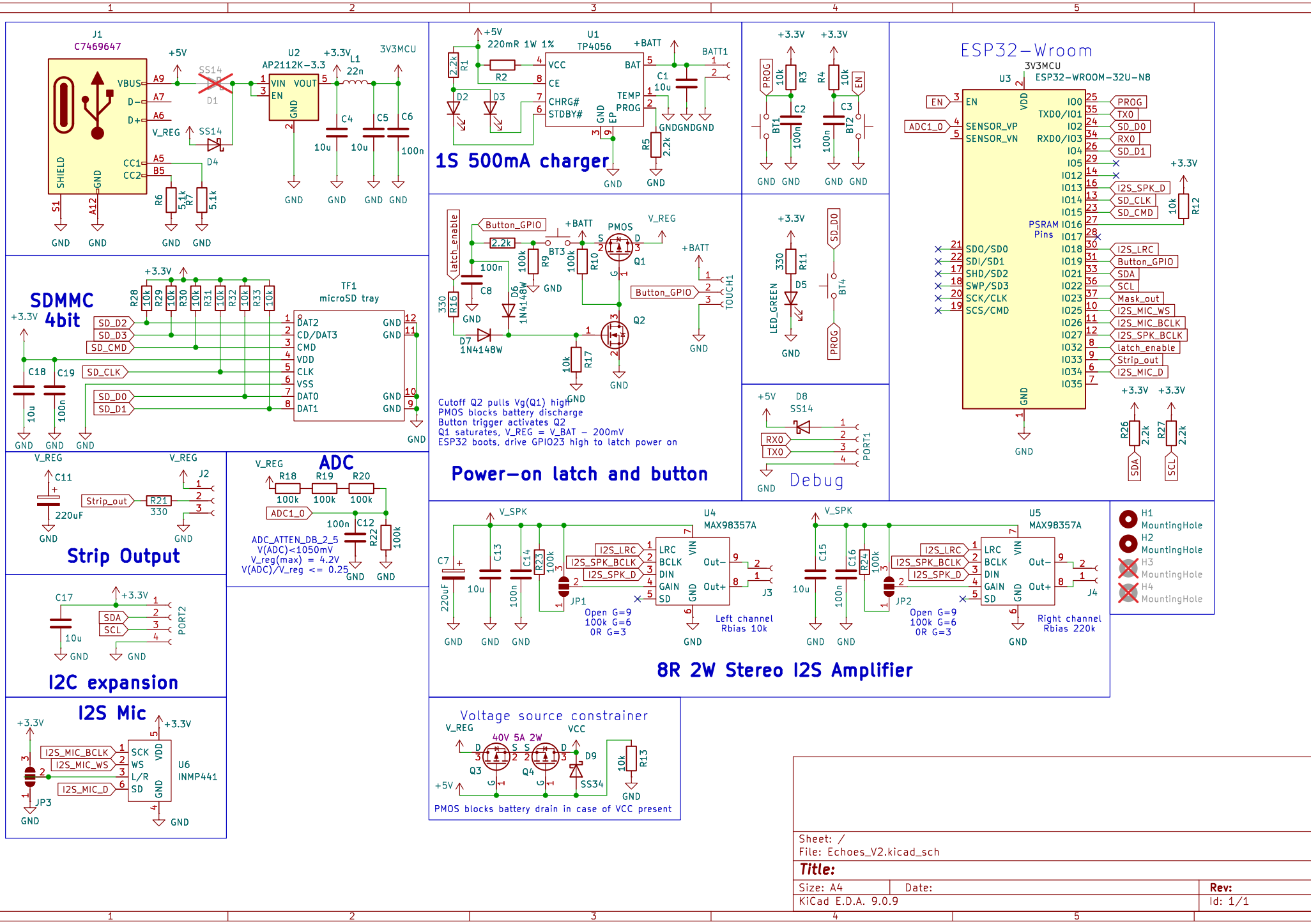

J1
C7469647
VBUS
D-
D+
V_REG
CC1
CC2
SHIELD
GND
S1
A12
SS14
D1
D4
R6
5.1k
R7
5.1k
U2
AP2112K-3.3
VIN
VOUT
EN
GND
+5V
+3.3V
L1
22n
3V3MCU
C4
10u
C5
10u
C6
100n
1S 500mA charger
U1
TP4056
220mR 1W 1%
R1
2.2k
R2
VCC
CE
CHRG#
STDBY#
BAT
TEMP
PROG
EP
+BATT
BATT1
C1
10u
D2
D3
R5
2.2k
PROG
10k
R3
R4
EN
C2
100n
C3
100n
BT1
BT2
ESP32-Wroom
U3
ESP32-WROOM-32U-N8
EN
ADC1_0
SENSOR_VP
SENSOR_VN
SDO/SD0
SDI/SD1
SHD/SD2
SWP/SD3
SCK/CLK
SCS/CMD
VDD
PSRAM Pins
IO0
TXD0/IO1
IO2
RXD0/IO3
IO4
IO5
IO12
IO13
IO14
IO15
IO16
IO17
IO18
IO19
IO21
IO22
IO23
IO25
IO26
IO27
IO32
IO33
IO34
IO35
PROG
TX0
SD_D0
RX0
SD_D1
I2S_SPK_D
SD_CLK
SD_CMD
I2S_LRC
Button_GPIO
SDA
SCL
Mask_out
I2S_MIC_WS
I2S_MIC_BCLK
I2S_SPK_BCLK
latch_enable
Strip_out
I2S_MIC_D
R12
10k
R26
2.2k
R27
2.2k
SDMMC 4bit
TF1
microSD tray
DAT2
CD/DAT3
CMD
VDD
CLK
VSS
DAT0
DAT1
SD_D2
SD_D3
SD_CMD
SD_CLK
SD_D0
SD_D1
R28
R29
R30
R31
R32
R33
C18
C19
10u
100n
Power-on latch and button
latch_enable
Button_GPIO
2.2k
R9
100k
R10
100k
BT3
PMOS
Q1
Q2
C8
100n
R16
330
D6
1N4148W
D7
1N4148W
R17
10k
TOUCH1
Cutoff Q2 pulls Vg(Q1) high
PMOS blocks battery discharge
Button trigger activates Q2
Q1 saturates, V_REG = V_BAT - 200mV
ESP32 boots, drive GPIO23 high to latch power on
LED_GREEN
R11
330
D5
BT4
D8
SS14
RX0
TX0
PORT1
Debug
Strip Output
C11
220uF
Strip_out
R21
330
J2
ADC
R18
R19
R20
100k
ADC1_0
C12
100n
R22
100k
ADC_ATTEN_DB_2_5
V(ADC)<1050mV
V_reg(max) = 4.2V
V(ADC)/V_reg <= 0.25
I2C expansion
C17
10u
SDA
SCL
PORT2
I2S Mic
U6
INMP441
SCK
WS
L/R
SD
I2S_MIC_BCLK
I2S_MIC_WS
I2S_MIC_D
JP3
8R 2W Stereo I2S Amplifier
U4
MAX98357A
U5
MAX98357A
V_SPK
C7
220uF
C13
10u
C14
100n
R23
100k
JP1
Open G=9
100k G=6
OR G=3
Left channel
Rbias 10k
J3
C15
10u
C16
100n
R24
100k
JP2
Right channel
Rbias 220k
J4
LRC
BCLK
DIN
GAIN
SD
Out-
Out+
VIN
Voltage source constrainer
40V 5A 2W
Q3
Q4
D9
SS34
R13
10k
PMOS blocks battery drain in case of VCC present
H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole
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File: Echoes_V2.kicad_sch
Title:
Size: A4
Date:
Rev:
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